

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT2-Constraint-Based Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS:3

Total Marks:30

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- D1 cannot work Night shift
- D2 must work before D3 (Morning < Afternoon < Night)
- Only one doctor per shift
- D3 cannot work Morning
- All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- Movement allowed: Up, Down, Left, Right
- Cost of each move = 1
- Cannot pass through obstacles
- Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information. The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- Analyze all the given constraints and explain how they affect the robot's decision-making process.
- Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type
- Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

I K.B.Devarshan/192511426 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible solutions satisfying	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5

	all constraints				
Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

ANSWERS:

1.Constraints(M-Morning,A-Afternoon,N-Night):

1. $D_1 \neq N$
2. $Time(D_2) < Time(D_3)$ (i.e., $M < A < N$)
3. $AllDiff(D_1, D_2, D_3)$ (Single doctor per shift)
4. $D_3 \neq M$

Step-by-Step Backtracking Search:

1. **Initial State:**

- Assignment: $\{\}$
- Domains based on unary constraints (C_1, C_4):
 - $D_1: \{M, A\}$
 - $D_2: \{M, A, N\}$
 - $D_3: \{A, N\}$

2. **Iteration 1: Assign D_1**

- Choose $D_1 = M$.
- Current Assignment: $\{D_1 = M\}$
- applying C_3 , removing Ma
 - $D_2: \{A, N\}$
 - $D_3: \{A, N\}$

3. **Iteration 2: Assign D_2**

- Choose $D_2 = A$.
- Current Assignment: $\{D_1 = M, D_2 = A\}$
- Update Remaining Domains (applying C_3 , removing A):
 - $D_3: \{N\}$
- Check Constraint C_2 ($D_2 < D_3$): Afternoon < Night (Valid).

4. **Iteration 3: Assign D_3**

- Choose $D_3 = N$.
- Current Assignment: $\{D_1 = M, D_2 = A, D_3 = N\}$
- Check all constraints:
 - $D_1 \neq N \rightarrow M \neq N$ (Pass)
 - $D_2 < D_3 \rightarrow A < N$ (Pass)
 - $D_3 \neq M \rightarrow N \neq M$ (Pass)
 - All diff used exactly once (Pass)

Final Valid Schedule:

- **Morning:** D_1
- **Afternoon:** D_2
- **Night:** D_3

2. Robot Grid Navigation

Problem Statement: Standard Breadth-First Search (BFS) is an *uninformed* search and does not use heuristics. Since the problem strictly requires the **Manhattan Distance heuristic**, I will apply **Best-First Search** --which is often what is intended when a hard thing is introduced to a standard queue-based algorithm. , because the exact coordinates of S, G, and the obstacles were omitted, I have defined a standard scenario to demonstrate the exact steps required.

Assumed Environment setup:

- Grid: 5x5 (Coordinates 0,0 to 4,4)
- Start (*S*): (0,0)
- Goal (*G*): (2,2)
- Obstacles: (1,0) and (1,1)
- Heuristic function: $h(n) = |x_n - x_g| + |y_n - y_g|$

Node Expansion & Priority Queue (Best-First Search):

Step	Current Node Expanded	Evaluated Neighbors & their $h(n)$	Priority Queue (Sorted by lowest h)
1	(Initial State)	Start (0,0), $h = 0 - 2 + 0 - 2 = 4$	$[(0,0, h = 4)]$
2	(0,0)	(0,1) $\rightarrow h = 3$. (1,0) is blocked	$[(0,1, h = 3)]$
3	(0,1)	(0,2) $\rightarrow h = 2$. (1,1) is blocked	$[(0,2, h = 2)]$
4	(0,2)	(1,2) $\rightarrow h = 1$	$[(1,2, h = 1)]$
5	(1,2)	(2,2) $\rightarrow h = 0$	$[(2,2, h = 0)]$
6	(2,2)	Goal Reached	Empty

- **Optimal Path:** (0,0) $\rightarrow$ (0,1) $\rightarrow$ (0,2) $\rightarrow$ (1,2) $\rightarrow$ (2,2)
- **Total Cost:** 4 moves $\times$ 1 cost/move = 4

3. Autonomous Rescue Robot Scenario

a) Constraint Analysis & Decision Making The constraints fundamentally shift this from a simple pathfinding problem to a complex exploration problem:

- **Limited Sensing:** Because the robot only sees adjacent cells, it cannot calculate a complete path to the goal offline. It must interleave planning and execution.
- **Varying Costs:** The existence of risky zones (cost = 3) alongside normal zones (cost = 1) means the robot cannot just use a simple step-count (like BFS); it must actively weigh the cost of traversing a known risky path versus the potential cost of exploring an unknown path.
- **Obstacles:** Finding blocked paths mid-route forces the robot to backtrack or recalculate, making memory of past states critical to avoid infinite loops.

b) Solution Approach (Online Search Strategy)

The most appropriate strategy is an **Online Uniform Cost Search** paired with a dynamically updating map (similar to Real-Time A*).

1. **Initialize:** The robot starts at S and initializes an internal "map" with all cells marked as *unknown*.
2. **Perceive:** The sensors scan adjacent cells (Up, Down, Left, Right). It updates its internal map with found obstacles, risky zones, and safe zones.
3. **Plan:** Using the updated local map, the robot calculates the lowest-cost path to the closest unexplored boundary that leads toward G (cost: safe = 1, risky = 3).
4. **Act:** The robot executes the first step of this plan.
5. **Repeat:** After moving, it returns to step 2, continually looping until G is reached.

c) Application of AI Concepts

- **Intelligent Agent:** The robot acts as an agent by utilizing sensors to perceive the environment, processing this through an internal state model, and utilizing actuators to move (Up/Down/Left/Right) toward the goal.
- **Rationality:** The agent demonstrates rationality by always choosing the action that maximizes its expected performance measure-specifically, mathematically minimizing the total path cost based *only* on the percept sequence it has gathered so far.
- **Environment Type:** The environment is **Partially Observable, Deterministic, Sequential, Dynamic**

d) Justification of Approach

- An offline search algorithm (Dijkstra's) would fail completely here, they require the entire map to be known before the first move is made.
- An **Online Search Agent** is strictly necessary because the environment is partially observable. By dynamically updating its state map and evaluating uniform costs on the fly, the robot safely adapts to unexpected obstacles and penalizing risky zones, ensuring the survivor is reached with the actual minimized cost.